

**Applied Agentic AI for PMs/TPMs**

## **Capstone Project**

CalendarMate: AI-Powered Productivity Assistant

## Company Profile

ServionIQ Solutions is a SaaS-based platform delivering intelligent solutions tailored for professional services firms — consulting, legal, accounting, architecture, and marketing agencies. By blending domain expertise with cloud-native technologies, ServionIQ enables firms to efficiently manage projects, optimize resource allocation, and enhance client delivery.

The end-to-end platform integrates project management, time tracking, CRM, billing, workforce planning, and knowledge management — empowering firms to scale with agility, transparency, and precision.

## Your Role

You have been recently hired by the CTO as an AI PM/TPM. You lead the development and delivery of AI-powered solutions that drive real-world impact, with a strong focus on emerging technologies like Agentic AI.

Your role involves translating complex business needs into technically feasible product strategies, collaborating closely with cross-functional teams including machine learning engineers, data scientists, and designers.

## Problem Statement

ServionIQ’s PMs/TPMs and senior ICs manage a heavy operational load. A typical day involves:

- 8 to 14 meetings across engineering, product, design, and client teams
- 3 to 5 different team calendars that need to be cross-referenced before scheduling anything
- 30 to 45 minutes every morning figuring out what is happening today and what to prepare for
- Frequent double-bookings and conflicts because availability checks are manual
- Missed follow-ups: action items from meetings are discussed but rarely tracked systematically
- 60+ emails per day, many of which are meeting-related (invites, reschedules, FYIs, follow-ups)
- No consolidated view of the day — people switch between calendar, email, Slack, and docs

The result: PMs/TPMs spend roughly 2 hours per day on meeting logistics, scheduling, and email triage instead of high-value work. The CTO has asked you to build CalendarMate — an AI-powered productivity assistant that reduces this operational overhead.

## Curriculum Concepts You Will Apply

| Concept                   | Mandatory / Extended | How It Applies in CalendarMate                                                                                |
|---------------------------|----------------------|---------------------------------------------------------------------------------------------------------------|
| Multi-Agent System Design | Mandatory            | Design agents with clear responsibilities (briefing, scheduling, email, follow-up) and defined inputs/outputs |

|                            |                     |                                                                                                           |
|----------------------------|---------------------|-----------------------------------------------------------------------------------------------------------|
| Orchestration Pattern      | Mandatory           | Select and document why you chose sequential, router, or hierarchical orchestration for your agent system |
| Prompt Engineering         | Mandatory           | Write system prompts with grounding rules that prevent inventing meetings or fabricating attendee details |
| Tool Integration           | Mandatory           | Connect agents to Calendar API and Email API using native nodes or HTTP requests                          |
| Evaluation & Observability | Mandatory           | Connect to a tracing tool, run the provided test dataset, and document findings                           |
| Fine-Tuning                | Extended (optional) | Train a custom model on your evaluation data to improve specific agent behavior                           |
| RAG / Knowledge Retrieval  | Extended (optional) | Use retrieval to ground responses in actual calendar and email data                                       |
| Human-in-the-Loop          | Extended (optional) | Add approval steps where a PM/TPM reviews before creating events or sending emails                        |

## Orchestration Pattern Requirements

Your submission must document which orchestration pattern you selected and why. The three patterns from the course curriculum are:

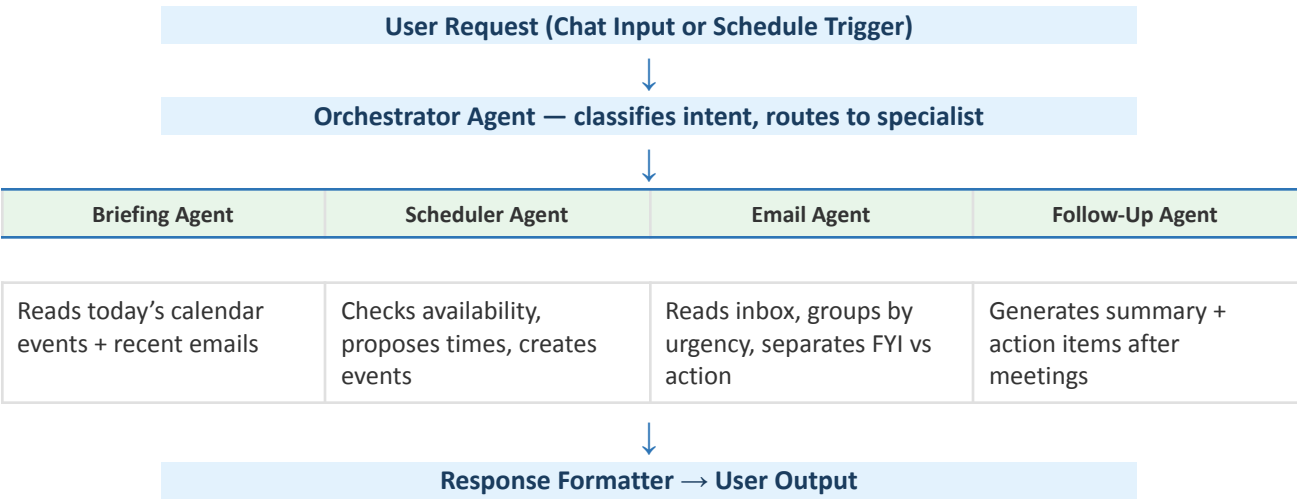
| Pattern             | When to Use                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| Sequential Pipeline | Each agent processes in a fixed order. Best when every request needs all agents in the same sequence.                  |
| Router / Dispatcher | A central orchestrator classifies the request and routes to the right specialist agent. Best when requests are varied. |
| Hierarchical        | An orchestrator decomposes complex requests into subtasks, delegates to specialist agents, and aggregates results.     |

### Recommendation for CalendarMate:

A Router pattern fits well because user requests can be very different: 'What's my day look like?' routes to a Briefing Agent, 'Schedule a meeting with John' routes to a Scheduler Agent, and 'Summarize my emails' routes to an Email Agent. However, you may choose any pattern and justify your decision.

## Suggested Architecture Overview

This is a reference architecture. You are free to design your own. Your submission must include your own architecture diagram.



Possible Agents

| Agent           | Role                                                                       |
|-----------------|----------------------------------------------------------------------------|
| Orchestrator    | Routes user requests to the right specialist based on intent               |
| Briefing Agent  | Pulls calendar events and emails, generates daily summary with conflicts   |
| Scheduler Agent | Checks availability, proposes slots, creates events with Google Meet links |
| Email Agent     | Reads and summarizes unread emails, groups by urgency/topic                |
| Follow-Up Agent | After meetings, generates action items and sends summaries to attendees    |

You may combine, split, or rename agents. You may use fewer or more agents than shown above.

What CalendarMate Should Be Able to Do

Core Capabilities (Required)

**Daily Briefing:** Generate a consolidated summary of the day: meetings with times and attendees, scheduling conflicts, priority emails requiring response, and a suggested focus plan.

**Smart Scheduling:** Accept natural language scheduling requests, check calendar availability, propose open time slots, and create the calendar event with a Google Meet link once confirmed.

**Email Summarization:** Summarize unread or recent emails, grouping them by urgency or topic. Separate action-required emails from FYI emails.

**Extended Capabilities (Choose at least one)**

**Meeting Follow-Up Generator:** After a meeting ends, generate a follow-up email with action items and send to attendees.

**Conflict Resolver:** Detect scheduling conflicts, assess priority, suggest rescheduling options.

**Attendee Research:** Look up external attendees before a meeting, provide a brief on their role and email history.

**Weekly Digest:** Weekly summary of meetings attended, action items, emails, and focus time vs meeting time.

**Evaluation Requirement**

Connect your CalendarMate workflow(s) to any evaluation/observability tool. You must:

- Run the provided baseline test dataset (below) through your pipeline
- Document the output for each test input
- Check whether each output contains the expected keywords/elements listed
- Test additional edge cases beyond the provided dataset
- Experiment with prompt modifications and document how output quality changes

**Baseline Test Dataset for Core Capabilities**

Run each of these through your pipeline. The 'Expected Output Must Contain' column tells you what a correct response should include. This is how TAs will evaluate your outputs.

| ID | Test Input                                     | Type     | Expected Output Must Contain                                                                 |
|----|------------------------------------------------|----------|----------------------------------------------------------------------------------------------|
| T1 | What does my day look like today?              | Briefing | List of today’s meetings with times and attendees. Must mention any conflicts if they exist. |
| T2 | Give me a summary of my meetings for tomorrow. | Briefing | Tomorrow’s meetings listed with times. Must NOT invent meetings.                             |

|     |                                                                                  |                             |                                                                                                                 |
|-----|----------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------|
| T3  | Schedule a 30-minute meeting with sarah@company.com tomorrow at 2pm.             | Scheduling                  | Must check availability. Must create event with Google Meet link. Must confirm time and attendee.               |
| T4  | Set up a sync with the design team sometime this week.                           | Scheduling (ambiguous )     | Must ask for clarification: which day, what time, who specifically. Must NOT create event without confirmation. |
| T5  | Find a time for a 1-hour meeting with john@ext.com and lisa@ext.com next Monday. | Scheduling (multi-attendee) | Must check availability for all participants. Must propose slots, not just pick one.                            |
| T6  | Schedule something for Sunday at 6am.                                            | Scheduling (edge case)      | Must flag that this is a weekend and unusual hour. Should ask for confirmation or warn.                         |
| T7  | Summarize my unread emails.                                                      | Email                       | Must group by urgency or topic. Must separate action-required from FYI. Must NOT invent email content.          |
| T8  | What emails need my attention today?                                             | Email                       | Must identify action-required emails only. Must include sender and subject. Must NOT fabricate emails.          |
| T9  | Book a meeting but I forgot with whom.                                           | Scheduling (missing info)   | Must ask who the meeting is with. Must NOT create an event with missing attendee.                               |
| T10 | Do I have any conflicts this week?                                               | Briefing                    | Must check all week's events for overlapping times. Must list specific conflicts or confirm none exist.         |
| T11 | Schedule a meeting at the same time as my existing standup.                      | Scheduling (conflict)       | Must detect the conflict. Must warn user. Must suggest alternative times.                                       |
| T12 | What's happening next Thursday?                                                  | Briefing                    | Must show events for that specific day. Must NOT show wrong day's events.                                       |

### Fine-Tuning (Optional Enhancement):

If you want to go further, use your evaluation data to fine-tune one or more agents. Not required but will be recognized in scoring.

## Tools and Technologies

| Category                 | What You Need                                    | Examples (not exhaustive)                          |
|--------------------------|--------------------------------------------------|----------------------------------------------------|
| Workflow/Agent Platform  | Low-code or no-code platform                     | n8n, LangFlow, or any platform                     |
| Calendar API             | Read events, check availability, create meetings | Google Calendar API, Outlook Calendar API          |
| Email API                | Read, search, and send emails                    | Gmail API, Outlook Mail API                        |
| LLM                      | Reasoning, summarization, NLU                    | OpenAI (gpt-4.1-mini, gpt-4o-mini), Claude, Gemini |
| Observability/Evaluation | Monitor traces, track quality                    | LangWatch, Langfuse, or any tracing tool           |
| Authentication           | Secure API access                                | OAuth2 (Google, Microsoft)                         |

## Assignments

### Q1: Business Case (15 points)

Identify and document the business benefits of implementing Agentic AI to manage employee productivity at ServionIQ. Include at least 3 specific metrics you would track.

### Q2: Build CalendarMate (45 points)

Build the CalendarMate AI assistant. Your submission must include:

**Design and Architecture:** Architecture diagram, functionality mapping, tool selection rationale, agent design decisions, orchestration pattern selection with justification.

**Implementation:** Core + extended capabilities, observability integration, testing with baseline dataset + edge cases.

**Cost and Evaluation Strategy:** Cost per user per day, 3+ production metrics, testing observations.

| Sub-criteria                                                          | Points |
|-----------------------------------------------------------------------|--------|
| Design: architecture diagram + orchestration pattern + tool rationale | 10     |
| Core capabilities working end-to-end                                  | 12     |

|                                                 |   |
|-------------------------------------------------|---|
| Extended capability implemented                 | 8 |
| Observability connected with traces             | 5 |
| Baseline dataset tested with documented results | 5 |
| Cost analysis and evaluation strategy           | 5 |

### Q3: Program Charter (25 points)

Create the program charter: vision, scope, success criteria, timeline, risks, stakeholders, decision-making, rollout plan.

### Q4: Reflection and Next Steps (15 points)

1-page reflection: trace findings, 2-week improvement plan, privacy/security considerations, rollout plan for 50 PMs/TPMs, connection to evaluation/fine-tuning skills.

## Submission Guidelines

- Export workflow(s) as JSON, upload to GitHub repository with README
- Screenshots: workflow canvas, CalendarMate in action, observability traces
- Architecture diagram (Excalidraw, draw.io, Miro, PowerPoint, or hand-drawn)
- Baseline test dataset results documented (all 12 inputs with outputs)
- Submit GitHub repository link along with documents